Design Verification Plan and Report (virtual test version) — VBF-T4R1-DVPR-01

One row per verification item; result values from simulation outputs; uncovered conditions honestly marked N/A.

Item	Condition	Result value	Determination	Source
1C discharge capacity	1C constant-current discharge, 25 °C, 2.5 V cutoff	5.0368 Ah	pass (vs nominal 5.0 Ah: 100.7 %)	cell/r6_t2_1c_dfn.json:capacity_ah
–20 °C 1C capacity retention	1C discharge, 253.15 K cold-soaked start, h = 80 (self-heating removed)	0.99267	pass (≥ 0.95)	cell/r6_t2_retention.json:lowT_retention
Energy density	contract formula (∫V·I dt / Σ layer thickness×(1–ε)×density×area)	471.55 Wh/kg	pass (≥ 327.18)	cell/r6_t2_energy.json:energy_density_wh_kg
Volumetric energy density	energy ÷ stack volume (188.8 µm × 0.1027 m²)	925.77 Wh/L	pass (≥ 880)	cell/r6_t2_energy.json:energy_density_wh_l
4C fast-charge temperature rise	4C charge after 1C discharge to 2.5 V, 45 °C ambient, lumped thermal	T_max 327.70 K (+9.6 K above ambient)	pass (≤ 333.15 K)	cell/r6_t2_4c45C_dfn.json:T_max_K
4C fast-charge plating	same protocol, plating submodel	anode potential min +0.0205 V (≥ 0 V)	pass (no plating)	cell/r6_t2_4c45C_dfn.json:anode_potential_v
Voltage window	parameter-set cut-offs	2.5 – 4.2 V	pass (per design)	Chen2020 parameter set
DC resistance	calc-energy DCR caliber	4.06 mΩ	informational (no threshold)	cell/r6_t2_energy.json:dcr_ohm

Items explicitly N/A (beyond pure simulation boundary, require physical experiment): nail penetration; overcharge to thermal runaway; crush; drop; cycle life (no aging model); rate-pulse internal resistance.

Conclusion

All seven simulation-based verification items pass against the entry-0 criteria. Uncovered items (mechanical abuse, cycle life, pulse resistance) are outside the pure-simulation boundary and require physical prototypes — listed above for the paper's limitations section. The –20 °C retention value is computed under the flat-transport design-target profile (LHCE-class idealization); physical confirmation of electrolyte conductivity at –20 °C is recommended.